

Arithmetic Integers

Review Exercise Questions

Level-2

- Q1.** For any integer n , prove that the number $n^5 - n$ will always be divisible by 30.
- Q2.** If n is an odd integer, the number $n^4 - 1$ must be a multiple of 32. Is this true or false?
- Q3.** Prove that $n^5 + 4n$ is divisible by 5 for any integer n .
- Q4.** Two integers a, b leave remainders of r_1, r_2 when divided by n . Show that:
- A. The remainder of ka when divided by n will be the same as the remainder of kr_1 when divided by n (k is any integer).
 - B. The remainder of b^2 when divided by n will be the same as the remainder of r_2^2 when divided by n .
 - C. The remainder of $a + b$ when divided by n will be the same as the remainder of $r_1 + r_2$ when divided by n .
 - D. The remainder of $a - b$ when divided by n will be the same as the remainder of $r_1 - r_2$ when divided by n .
 - E. The remainder of ab when divided by n will be the same as the remainder of $r_1 r_2$ when divided by n .
- Q5.** (a) Prove that $n^2 + 1$ cannot be a multiple of 3.
(b) Prove that $n^3 + 2$ cannot be a multiple of 9.
(c) Prove that if n is an odd integer, then $n^3 - n$ must be a multiple of 24.
(d) Prove that $n^3 + 2n$ is always divisible by 3.
- Q6.** Prove that the fraction $\frac{12n+1}{30n+2}$ is in the simplest form.
- Q7.** (a) Prove that m is a prime number greater than 3, then $m^2 - 1$ must be divisible by 24.
(b) Prove that if m and n are prime numbers greater than 3, then $m^2 - n^2$ must be divisible by 24.

Q8. Find the last digit of (a) 2^{100} (b) 77^{77} .

Q9. If a, b, c are three integers such that $a^2 + b^2 = c^2$, then prove that at least one of them must be divisible by 3.

Q10. x, y, z are three natural numbers whose sum is divisible by 6. Prove that $x^3 + y^3 + z^3$ is also divisible by 6.

